

★ WEEK 16 · PART 3 of 3

The Final Project, and the Course Close

From an Honest Comparison to a Trading Decision

The project, step by step · the competencies you carry · what's next

The finale. We turn the fair comparison into a deliverable: one asset, three families, a held-out test, and a verdict that becomes a trading recommendation. Then we close the core course and look ahead.

3

PART 3 · THE FINAL PROJECT

Comparison to Decision to Delivery

Your final project step by step, turning the winning model into a trading decision, presenting the evidence honestly, the three competencies you now hold, and where the journey goes next.

From Method to Deliverable

Parts 1-2 built it; Part 3 ships it

1

Part 1 gave us the choice

Three families, three forecast shapes - matched to the decision at hand.

2

Part 2 gave us the test

A fair, out-of-sample comparison against the naive benchmark, guarded against look-ahead bias.

3

Part 3 gives us the deliverable

One asset run end to end, an honest verdict, and a trading recommendation you can defend.

Part 3 in one line: turn the comparison into a decision, present it honestly, and close the course.

Your Final Project, Step by Step

One asset, three families, a fair test, a verdict

The capstone is a complete, honest study of one asset. You pick the series, split it cleanly, fit all three families on the past, grade them on the future, and end with a verdict that ties the winner to a real trading decision - caveats included.

What you produce

- A clean train/test (or walk-forward) setup.
- A forecast from each of the three families.
- A scoreboard and a decision-driven verdict.

What makes it good

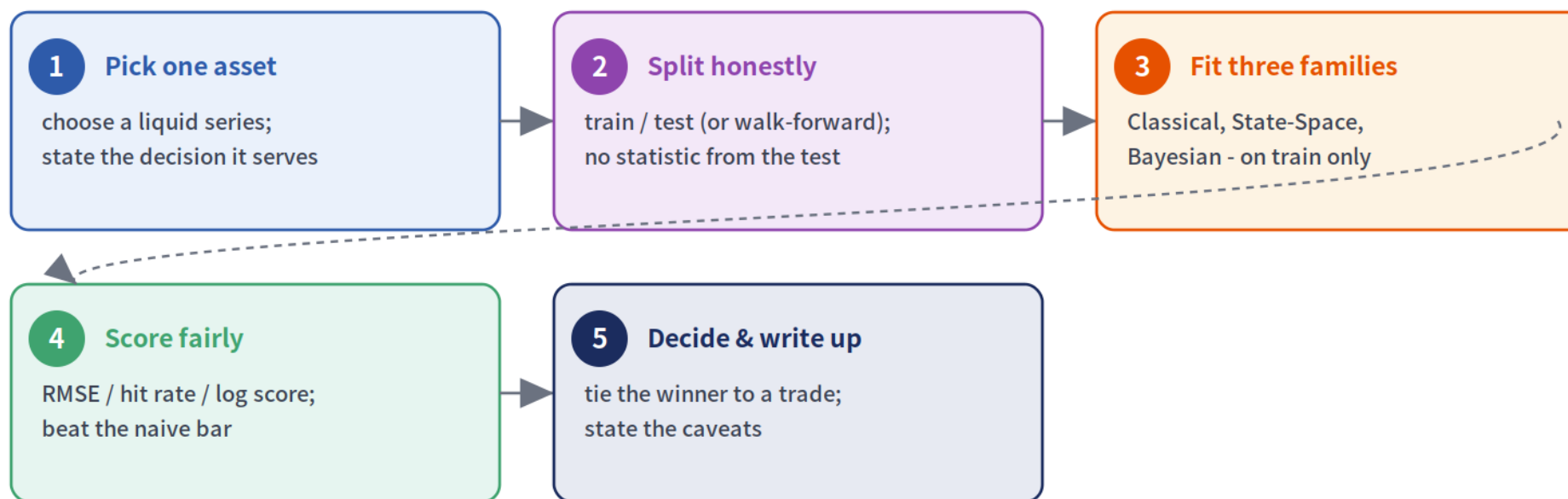
- Honesty: no leakage, real out-of-sample scores.
- Clarity: the decision the study serves is stated.
- Humility: the caveats are as visible as the result.

Your Final Project, Step by Step

Pick one asset and state the decision; split honestly; fit all three families on the training data; score them fairly against the naive bar; then decide and write it up. The deliverable is the verdict, not the model!

Your Final Project, Step by Step

One asset, three families, a fair test, an honest verdict - and a trading recommendation.



The deliverable is not the model - it is the honest verdict and the decision it supports.

Pick the Asset and the Decision

Start from what you're trying to do

Choosing the asset

- A liquid series with enough history.
- Something whose behavior you can reason about.
- The pair from your thread is a natural choice.

Stating the decision

- What will the forecast actually support?
- A trade signal, a hedge, or a risk limit?
- The decision sets which forecast shape wins.

Split Honestly, Fit Three Families

The disciplined middle of the project

Split honestly

- Reserve the most recent slice as the test.
- Consider walk-forward for a stronger grade.
- Compute no statistic on the test data.

Fit three families

- Classical, state-space, and Bayesian.
- All on the training period only.
- One clear representative from each.

Score Fairly, Then Decide

Turn the scoreboard into a verdict

Score fairly

- RMSE, hit rate, and a log score for the band.
- Every family against the naive benchmark.
- Read several metrics together, not one.

Then decide

- Pick the family whose output the decision needs.
- Break near-ties with simplicity.
- State the verdict in one honest sentence.

From Comparison to a Trading Decision

Tie the models back to the pairs strategy

A scoreboard is not the finish line - a decision is. The winning forecast feeds the pairs strategy: a point sets the hedge, a regime probability gates the trade, and a distribution sizes the risk. The final test is whether it improves the out-of-sample P&L.

How each output is used

- A point forecast sets the hedge ratio.
- A regime probability gates on/off and size.
- A distribution sets the risk and the stop.

The real verdict

- Does the signal beat buy-and-hold out of sample?
- Is the improvement worth the complexity?
- Would you actually trade it?

One Pair Trade, Every Model

This is where all the time series lands. Cointegration sets the hedge ratio and the spread; the z-score triggers the entry; OU gives the holding period; GARCH sizes it; the HMM gates it; and the Bayesian band sets the conviction. Six models, one trade ticket.

One Pair Trade, Every Model

The whole course, assembled into a single decision - each model contributes one number.



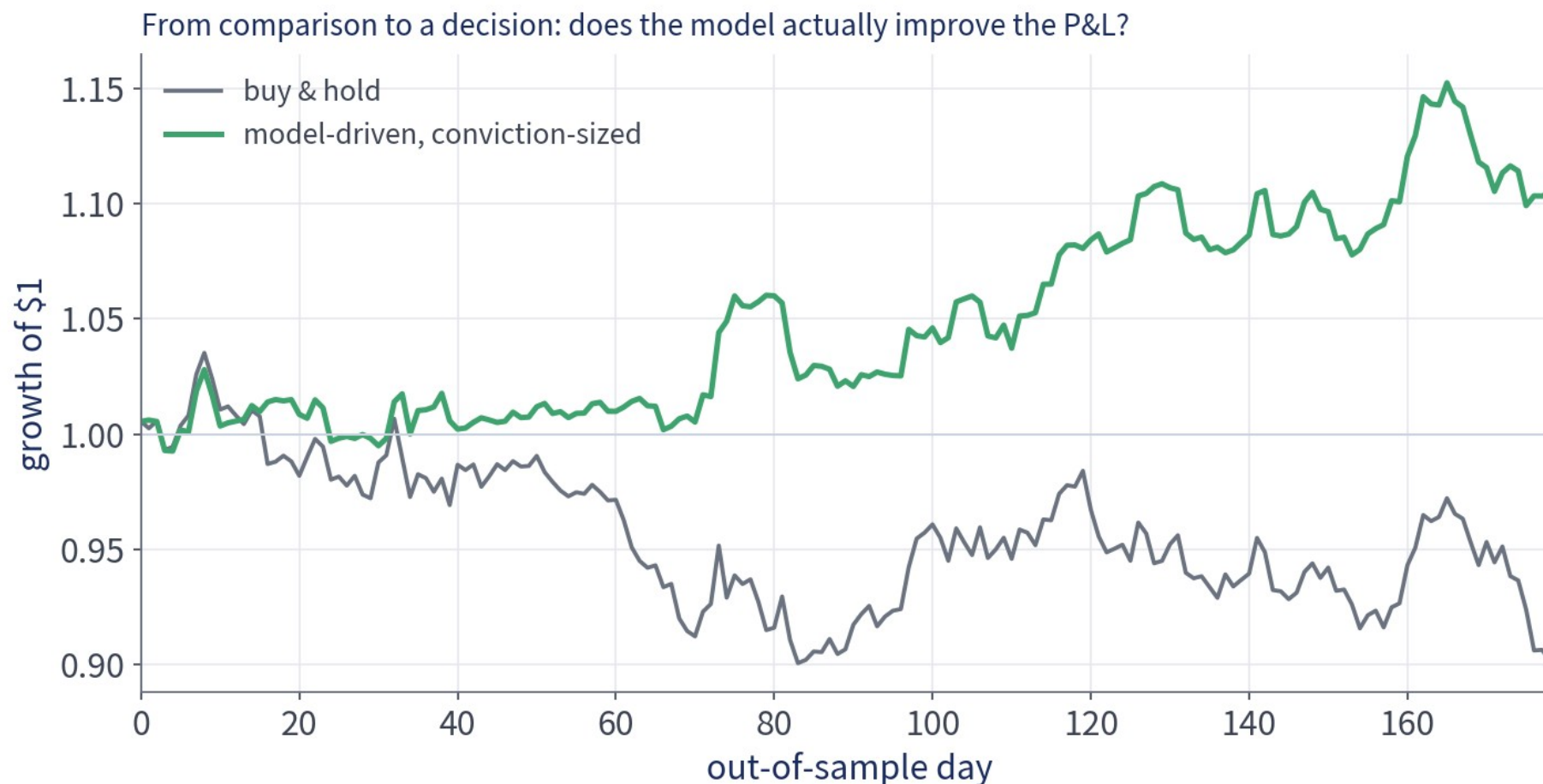
The trade ticket

SHORT the spread (short A, long 0.76 B), size 0.32 of the book, expected exit near $z = 0$ in about 2 days, stop out if P(crisis) rises above 0.5. Every number came from a model you built this course.

That is what all the time series was for.

Does the Model Improve the P&L?

The ultimate grade for a trading study: the model-driven, conviction-sized strategy against buy-and-hold, out of sample. A lower error only matters if it turns into a better, and honestly tested, equity curve.



Presenting It - and Pitfalls to Avoid

Make the evidence clear and the honesty obvious

A good capstone is judged as much on honesty as on results. Lead with the decision, show the fair test, and be upfront about the caveats. The most impressive thing you can present is a result you have genuinely tried to break.

How to present

- State the decision and the asset first.
- Show the split, the metrics, and the naive bar.
- End with a clear, one-line recommendation.

Pitfalls to avoid

- Hiding the caveats or the failed attempts.
- Over-claiming from a thin RMSE gap.
- A backtest you haven't stress-tested for leakage.

Three Competencies You Now Have

What you take out of the core course

Beyond any single model, you leave with three durable skills. You can match a model to a decision, you can test any claim honestly out of sample, and you can let uncertainty size the bet. These outlast any particular technique.

The three skills

- Match the model to the decision.
- Test every claim out of sample.
- Let honest uncertainty size the bet.

Why they last

- They apply to models we haven't met yet.
- They are how professionals avoid fooling themselves.
- They turn techniques into judgment.

Match the Model to the Decision

Start from the choice, not the tool

What it means

- Ask what the decision needs first.
- Pick the forecast shape that fits.
- Point, regime, or distribution.

Why it matters

- It stops you reaching for the fanciest tool.
- It keeps the work tied to a real purpose.
- It makes 'which model?' a solvable question.

Test Everything Out of Sample

The habit that keeps you honest

What it means

- Fit on the past, grade on the unseen future.
- Compare against a naive benchmark.
- Guard fiercely against look-ahead bias.

Why it matters

- It is how the field settled its arguments.
- It separates real edges from lucky fits.
- It is the difference between a demo and a strategy.

Let Uncertainty Size the Bet

The Bayesian habit, made practical

What it means

- Carry a credible band, not just a point.
- Scale exposure by conviction and by the band.
- Trade smaller when the model is unsure.

Why it matters

- It is the lesson of the quant quake.
- It turns honesty about doubt into safer risk.
- It connects the whole course to real trading.

What's Next: the Deep-Learning Special

Four bonus weeks - and the skepticism to keep

The core course ends here, but the journey continues. A deep-learning special covers neural forecasters and sequence models - powerful tools that deserve exactly the same scrutiny. Bring your out-of-sample discipline with you; it matters more, not less, as models grow

What's ahead

- Neural networks for time series.
- Sequence models and attention.
- When deep learning helps - and when it doesn't.

What to bring

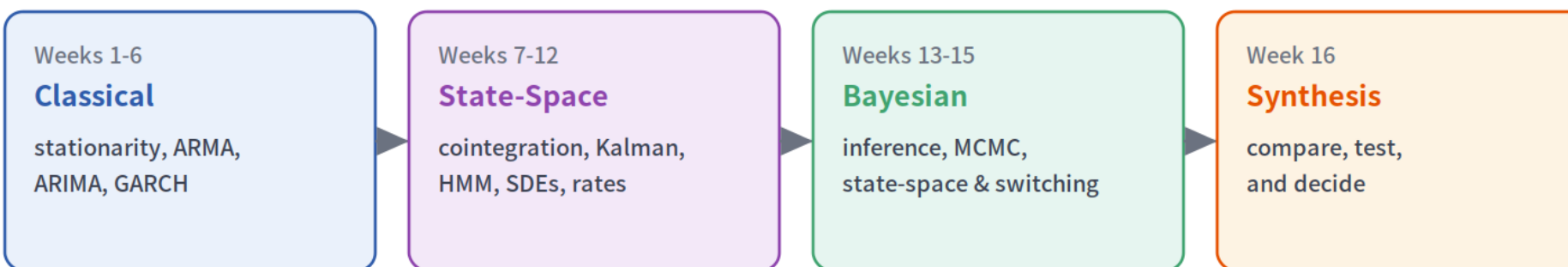
- The same fair, out-of-sample tests.
- The same naive benchmark.
- The same honesty about uncertainty.

The Core Course, Complete

Sixteen weeks across three families, one pairs thread running through them, and a deep-learning special ahead - to be held to the very same out-of-sample bar. Carry three things forward: decision, testing and honest uncertainty

The Core Course, Complete

Sixteen weeks, three families, one pairs thread - and the skepticism to carry forward.



What's next: the Deep-Learning special (four bonus weeks)

Neural forecasters and sequence models - powerful, and to be held to the very same out-of-sample bar.

Carry three things forward

Match the model to the decision; test everything out-of-sample; and let honest uncertainty size the bet.

Three Families on One Asset

Assemble the core of your capstone

GOAL Produce the full comparison for one asset, end to end, ready to write up.

[INPUT] Prompt · Claude Code

Write and run `capstone_compare.py` that:

1. loads one asset and makes an honest split;
 2. fits Classical, State-Space, and Bayesian models on the training data only;
 3. forecasts the test period with each family;
 4. scores RMSE, hit rate, and a log score against the naive benchmark, and prints a scoreboard;
 5. saves the forecasts and scores for the write-up.
- Flag any look-ahead risk it finds in the pipeline.

How to work with it

- One asset, one honest split.
- Three families, same test.
- Beat the naive bar.
- Keep the scoreboard for the report.

Deliverable: `capstone_compare.py` + a saved scoreboard and forecasts for one asset.

From Forecasts to a Trading Verdict

Turn the comparison into a recommendation

GOAL Convert the winning forecast into a trading decision and grade its out-of-sample P&L.

[INPUT] Prompt · Claude Code

Write and run `capstone_decision.py` that:

1. takes the winning family's forecast;
 2. turns it into a pairs-trade rule - a hedge from a point, a gate from a regime, a size from a band;
 3. simulates the strategy on the TEST period only;
 4. compares its equity curve and Sharpe to buy-and-hold;
 5. writes a one-paragraph verdict with caveats.
- State plainly whether you would trade it.

How to work with it

- Use only the test period.
- Compare to buy-and-hold.
- Report Sharpe, not just error.
- End with a clear recommendation.

Deliverable: `capstone_decision.py` + an out-of-sample equity curve and a written, caveated verdict.

Three Questions Before You Finish

If you can answer these, the course has landed

Q1 What is the real deliverable of the capstone?

**An honest verdict
and a decision**

Q2 What is the ultimate test of a trading study?

**Out-of-sample P&L vs
buy-and-hold**

Q3 What three competencies do you carry forward?

**Decision, testing,
honest uncertainty**

All three come from the project steps, the P&L verdict, and the three durable skills.

Three Pitfalls to Avoid

Where capstones go wrong

Common mistakes

- "A great scoreboard is the finished project."
- "Present only the version that worked."
- "The fanciest model must be the answer."

The reality

- The deliverable is the decision, not the score.
- Show the caveats and the failures too.
- Let the decision and the honest test decide.

Three Things to Remember

The whole week in one screen

1

Match the model to the decision

Three families answer three questions - point, regime, distribution. Start from what the decision needs, and there is no universal best model.

2

Test everything out of sample

Fit on the past, grade on the unseen future, beat the naive benchmark, and guard against look-ahead bias. The scoreboard, not taste, decides.

3

Deliver a decision, honestly

Turn the winning forecast into a trading recommendation, grade its out-of-sample P&L, and present the caveats as clearly as the result.

The core course is complete - three families, one thread, and the discipline to use them well.

Where You Go From Here

The core course is complete

You started with a single return series and a question about stationarity. You end able to model its mean and variance, track its hidden states, quantify its uncertainty, and test any claim about it honestly. That arc - from data to decision - is the whole point.

What you can now do

- Model returns with the classical family.
- Track hidden states and regimes.
- Quantify uncertainty the Bayesian way.

And crucially

- Choose among them for a real decision.
- Test every claim out of sample.
- Size the bet to honest uncertainty.

Keep the Skepticism

The habit worth more than any model

As models get fancier

- The temptation to trust them grows.
- So does the risk of fooling yourself.
- Deep learning is no exception.

So keep asking

- What decision does this serve?
- How does it do out of sample?
- How sure am I, really?

Three Families, One Thread, One Discipline

Advanced Financial Time Series

1

You learned three families

Classical for the mean and variance, state-space for hidden states and regimes, and Bayesian for honest, full uncertainty.

2

You followed one thread

The pairs strategy grew across sixteen weeks, from finding a pair to trading it with credible bands.

3

You keep one discipline

Match the model to the decision, test everything out of sample, and let honest uncertainty size the bet.

Thank you - and good luck with the final project.